Speculative Behavior with Bayesian Learning

Morris's (1996) Extension of the Harrison–Kreps Model

Thomas J. Sargent

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Outline

Overview

Structure of the Model

Information and Beliefs

Beta Priors and Posteriors

Market Prices with Learning

Two-Trader Analysis

Case A: Global Optimist

Case B: Perpetual Switching

Numerical Results

N–Trader Extension

Rate Dominance & Bubbles

Concluding Remarks

Overview

Overview

- Morris (1996) extended the Harrison–Kreps (1978) model of speculative asset pricing.
- Harrison–Kreps: risk-neutral traders with **dogmatic**, hard-wired heterogeneous beliefs about dividends.
- Morris: traders share the **same set of statistical models** but have **different prior distributions** over the parameter indexing those models.
- Traders update beliefs using **Bayes' Law** as dividend data arrive.

Key result: Before posterior distributions merge, traders disagree about the predictive density for dividends and therefore about the asset's value, generating *speculative premia*.

Key Features

1. All traders share one family of statistical models for dividends.
2. A single parameter θ indexes the family.
3. All traders observe the same dividend history.
4. All traders update via Bayes' Law.
5. Traders have different initial *prior* distributions over θ.
6. Posteriors eventually merge (consistency of Bayesian learning).
7. **Before** merging, traders disagree about the predictive density for dividends—hence about the asset's value.

Speculative Behavior

As in Harrison–Kreps, heterogeneous beliefs induce investors to engage in **speculative behavior**:

> *Sometimes a trader is willing to pay* **more** *for the asset than the trader's own "fundamental" value—the expected discounted value of the prospective dividend stream—because the trader anticipates being able to* **resell** *the asset to a more optimistic trader in the future.*

Short-sales are **prohibited**: pessimists can only express their views by selling shares, not by borrowing and selling.

Structure of the Model

Dividend Process

- Fixed supply of shares of an asset.
- Each share yields a stream of **binary** i.i.d. dividends $\{d_t\}$:

$$d_{t+1} \in \{0, 1\}.$$

- The dividend equals 1 with *unknown* probability $\theta \in (0, 1)$ and 0 with probability $1 - \theta$.

Contrast with Harrison–Kreps: there, traders had hard-wired beliefs about a Markov transition matrix. Here, θ is unknown and traders *learn*.

Traders and Timing

- A finite set $\mathcal{I}$ of **risk-neutral** traders.
- Common discount factor $\beta = 1/(1+r) \in (0,1)$.
- Asset is traded *ex dividend* each period $t = 0, 1, 2, \ldots$.
- Owner at end of period t receives d_{t+1} and may sell at $t+1$.
- **Short sales prohibited.**
- All traders have sufficient wealth to buy.

Trader i's **fundamental value** of the asset:

$$\sum_{j=1}^{\infty} \beta^j \, \hat{\theta}_i \;=\; \frac{\hat{\theta}_i}{r},$$

where $\hat{\theta}_i$ is the mean of trader i's posterior over θ.

Information and Beliefs

Bayesian Updating

At time $t \geq 1$, all traders observe $(d_1, \ldots, d_t)$.

All traders update via **Bayes' rule**.

Heterogeneous *priors* $\Rightarrow$ heterogeneous *posteriors* $\Rightarrow$ disagreement about the value of the asset.

Note: Morris's traders do *not* use data on past *prices* of the asset to update beliefs—only dividends.

Departure from the Harsanyi Doctrine

The **Harsanyi Common Prior Doctrine** (Harsanyi, 1967–68):

> *If two rational agents have the same information and reasoning capabilities, they will share the same joint probability distribution.*

- Harrison–Kreps departed radically (dogmatic beliefs).
- Morris departs **less** completely:
  - Agents *share* the same family of models.
  - They differ only in *initial priors* over the parameter θ.
- Defense: different "uninformative" priors are each defensible (e.g. uniform vs. Jeffreys).
- Motivation: Miller's (1977) "hot issue" anomaly for IPOs, where traders have little data about a new enterprise.

Beta Priors and Posteriors

Beta Prior Distributions

Trader i has a **Beta** prior:

$$\theta \sim \text{Beta}(a_i, b_i), \quad a_i, b_i > 0.$$

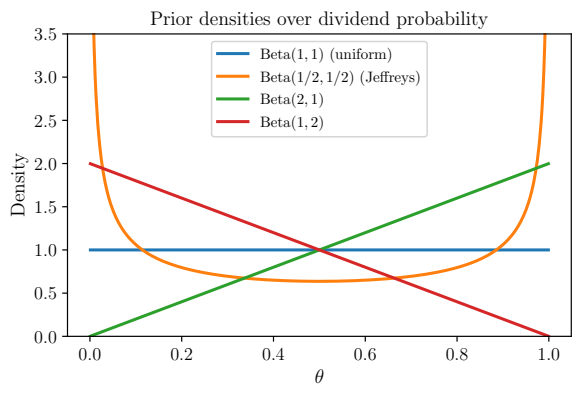

Figure 1: Selected Beta densities over the dividend probability θ

Conjugate Updating

After t periods with s dividend payments (successes):

$$\text{Prior: } \mathrm{Beta}(a_i, b_i) \xrightarrow{s \text{ successes in } t \text{ trials}} \text{Posterior: } \mathrm{Beta}(a_i + s,\ b_i + t - s).$$

Posterior mean (trader i's estimate of θ):

$$\mu_i(s,t) \;=\; \mathbb{E}[\theta \mid s,t] \;=\; \frac{a_i + s}{a_i + b_i + t}.$$

Morris calls $\mu_i(s,t)$ trader i's **fundamental valuation** after history (s,t).

It is the probability trader i assigns to receiving a dividend next period.

Market Prices with Learning

Most Optimistic Valuation

After history (s, t), the **most optimistic fundamental valuation** is:

$$\mu^*(s, t) \;=\; \max_{i \in \mathcal{I}} \; \mu_i(s, t).$$

The asset is held by the trader who assigns the *highest* value.

An owner has the **option to sell** after receiving the period's dividend—this option is priced into the equilibrium.

Equilibrium Price

The competitive equilibrium price (in current dollars) satisfies:

$$\tilde{p}(s,t,r) = \frac{1}{1+r}\Big[\mu^*(s,t)\big\{1+\tilde{p}(s{+}1,t{+}1,r)\big\} + \big(1-\mu^*(s,t)\big)\,\tilde{p}(s,t{+}1,r)\Big].$$

Normalized price:

$$p(s,t,r) = r\,\tilde{p}(s,t,r).$$

This is the price of the risky asset in terms of the riskless asset (whose price is $1/r$).

Normalized Price Recursion

The normalized price satisfies:

$$p(s,t,r) \;=\; \frac{r}{1+r}\,\mu^*(s,t) + \frac{1}{1+r}\Big[\mu^*(s,t)\,p(s{+}1,t{+}1,r) + \big(1-\mu^*(s,t)\big)\,p(s,t{+}1,r)\Big].$$

Equivalently:

$$p(s,t,r) = \mu^*(s,t) + \frac{r}{1+r}\Big[\mu^*(s,t)\,p(s{+}1,t{+}1,r) + \big(1-\mu^*(s,t)\big)\,p(s,t{+}1,r) - \mu^*(s,t)\Big].$$

Computed via **value-function iteration**: set $p^0 \equiv 0$ and iterate

$$p^{n+1}(s,t,r) = \frac{r}{1+r}\,\mu^*(s,t) + \frac{1}{1+r}\Big[\mu^*(s,t)\,p^n(s{+}1,t{+}1,r) + \big(1-\mu^*(s,t)\big)\,p^n(s,t{+}1,r)\Big].$$

Speculative Premium

When the identity of the most optimistic trader can *switch* with future dividend realizations:

$$p(s,t,r) > \mu_i(s,t) \quad \text{for } \textbf{all } i \in \mathcal{I}.$$

The **speculative premium**:

$$p(s,t,r) - \mu^*(s,t) > 0.$$

- The premium reflects the **option value** of reselling to whichever trader becomes temporarily more optimistic as dividends arrive.
- It exists whenever there is *no global optimist*.

Two-Trader Analysis

Rate Dominance

Definition: Rate Dominance
Trader 1 **rate-dominates** trader 2 if

$$a_1 \geq a_2 \quad \text{and} \quad b_1 \leq b_2.$$

Theorem: Global Optimist

1. If trader 1 rate-dominates trader 2, then $\mu_1(s,t) \geq \mu_2(s,t)$ for all (s,t). Trader 1 is a **global optimist**.
2. In this case $p(s,t,r) = \mu_1(s,t)$ and there is **no speculative premium**.

When *neither* trader rate-dominates the other, the identity of the most optimistic trader can switch $\Rightarrow$ speculative premium.

Posterior Means: Crossing vs. Non-Crossing

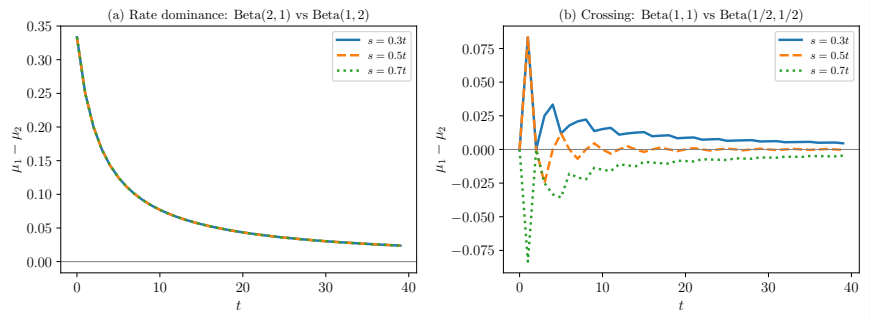

Figure 2: (a) Rate-dominant priors: $\mu_1 - \mu_2 > 0$ everywhere. (b) Non-dominant priors: posterior means cross, enabling speculative premia.

Case A: Global Optimist

Case A: Rate Dominance (No Premium)

Priors: Trader 1 $\sim$ Beta$(2,1)$, Trader 2 $\sim$ Beta$(1,2)$.

Trader 1 rate-dominates ($a_1 \geq a_2$, $b_1 \leq b_2$).

Quantity	**Value at** $(s,t) = (0,0)$
$\tilde{p}(0,0)$	2.0
Fundamental value, trader 1	2.0
Fundamental value, trader 2	1.0

The price **exactly equals** trader 1's perpetuity value.

No speculative premium.

Case B: Perpetual Switching

Case B: Perpetual Switching (Speculative Premium)

Priors: Trader 1 $\sim$ Beta$(1,1)$ (uniform), Trader 2 $\sim$ Beta$(1/2,1/2)$ (Jeffreys).

Neither trader rate-dominates the other.

Quantity	**Value at** $(s,t)=(0,0)$
$\tilde{p}(0,0)$	≈ 1.636
Fundamental value, trader 1	1.5
Fundamental value, trader 2	1.5

The price **exceeds both** traders' fundamental valuations.

The premium reflects the **option value** of reselling to whichever trader becomes temporarily more optimistic.

Numerical Results

Normalized Price vs. Interest Rate

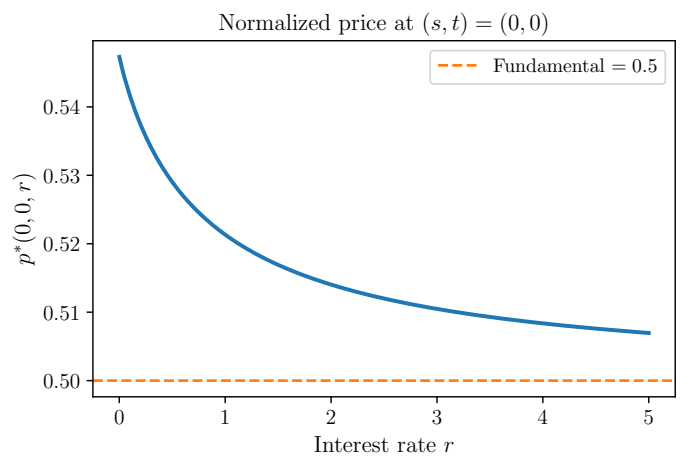

Figure 3: $p^*(0,0,r)$ as a function of r, for Beta$(1,1)$ vs. Beta$(1/2,1/2)$.

Normalized Price vs. Time

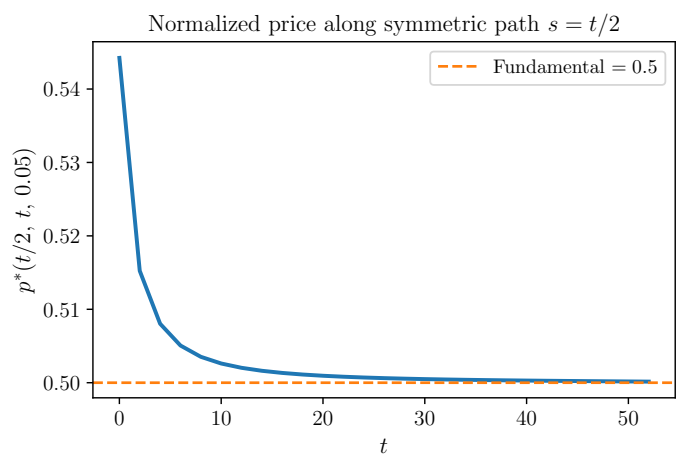

Figure 4: $p^*(t/2,\ t,\ 0.05)$ along the symmetric path $s = t/2$.

Speculative Premium Heat Map

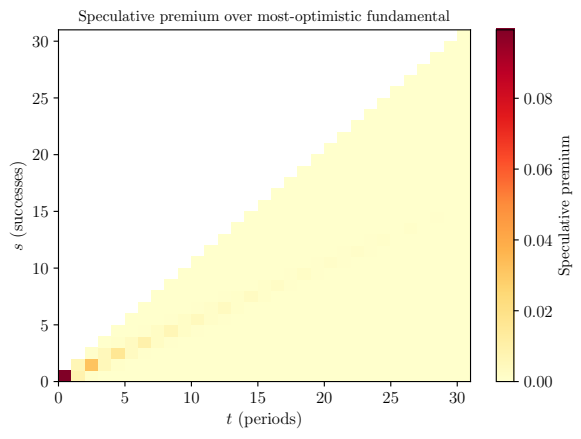

Figure 5: Speculative premium over the most-optimistic fundamental across histories (s, t), $\beta = 0.75$.

N–Trader Extension

General N–Trader Case

The recursion extends to any finite set of Beta priors $\{(a_i, b_i)\}_{i=1}^N$ by taking a max over all i at each period:

$$\tilde{p}(s,t) = \beta \max_{i=1,\dots,N} \Big[\mu_i(s,t)\big\{1 + \tilde{p}(s{+}1, t{+}1)\big\} + \big(1 - \mu_i(s,t)\big)\, \tilde{p}(s, t{+}1)\Big].$$

Three-trader example at $(s,t) = (0,0)$, $\beta = 0.75$:

	Prior	Fundamental value
Trader 1	$\text{Beta}(1,1)$	1.5
Trader 2	$\text{Beta}(1/2, 1/2)$	1.5
Trader 3	$\text{Beta}(3,2)$	1.8
Asset price $\tilde{p}(0,0)$		≈ 1.96

Rate Dominance & Bubbles

When Do Speculative Premia Arise?

Morris (1996) provides a **sharp characterization**:

Key Condition
There is a **speculative premium** if and only if there is **no global optimist**—i.e., no trader rate-dominates all others.

Examples:

Priors	Rate dominance?	Premium?
$\text{Beta}(2,1)$ vs. $\text{Beta}(1,2)$	Yes (Trader 1)	No
$\text{Beta}(1,1)$ vs. $\text{Beta}(1/2,1/2)$	No	Yes
$\text{Beta}(3,1)$, $\text{Beta}(2,1)$, $\text{Beta}(1,2)$	Yes (Trader 1)	No
$\text{Beta}(1,1)$, $\text{Beta}(1/2,1/2)$, $\text{Beta}(3/2,3/2)$	No	Yes

Checking Rate Dominance

Trader i **rate-dominates** all others iff

$$a_i \geq a_j \quad \text{and} \quad b_i \leq b_j \qquad \forall\, j \neq i.$$

- When this holds, $\mu_i(s,t) \geq \mu_j(s,t)$ for every history (s,t).
- Trader i is *always* the most optimistic $\Rightarrow$ no switching $\Rightarrow$ no premium.
- When it fails for every i, switching occurs along some histories $\Rightarrow$ speculative bubble.

Concluding Remarks

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- Morris (1996) enriches Harrison–Kreps (1978) by replacing dogmatic beliefs with **Bayesian learning**.
- Speculative premia arise naturally when traders express "maximal ignorance" differently (e.g. uniform vs. Jeffreys prior).
- The premium **declines over time** as posteriors merge—consistent with Miller's (1977) observation that IPO overpricing fades.
- Key mechanism: the **option to resell** to a future optimist raises the price above *every* trader's fundamental valuation.
- The no-short-sales constraint is essential: if pessimists could short freely, the bubble would be arbitraged away.

References

Harrison, J. M. and D. M. Kreps (1978).
Speculative investor behavior in a stock market with heterogeneous expectations.
Quarterly Journal of Economics, 92(2), 323–336.

Morris, S. (1996).
Speculative investor behavior and learning.
Quarterly Journal of Economics, 111(4), 1111–1133.

Miller, E. M. (1977).
Risk, uncertainty, and divergence of opinion.
Journal of Finance, 32(4), 1151–1168.

Harsanyi, J. C. (1967–68).
Games with incomplete information played by "Bayesian" players, Parts I–III.